

**Knowing Eye: A Web-Based Examination Platform with Behavior Monitoring
Using Facial and Postural Analysis**



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INTRODUCTION

Background of the Study

The use of digital technologies has changed how institutions conduct examinations, certifications, and competency assessments. Online and computer-based examination systems are now widely used by educational institutions, government agencies, and training organizations because they offer efficiency, scalability, and accessibility. However, as examinations shift to digital platforms, maintaining integrity, fairness, and reliability has become a major concern. Unlike traditional in-person exams, online assessments make it more difficult to monitor examinee behavior and prevent dishonest practices.

Globally, web-based examination systems support automated test administration, centralized data management, and remote access. However, existing studies identify one of the main limitations of online assessments as the difficulty of ensuring academic integrity when supervision is conducted remotely. Human proctors cannot effectively monitor a large number of examinees at the same time, especially in virtual environments. Because of this, researchers have explored the use of artificial intelligence and computer vision to improve monitoring through automated detection of behavior and identity verification.

Recent studies show the growing use of AI-based proctoring systems in online examinations. The iExam system [1] utilizes face detection and recognition to monitor students, achieving 90.4% face detection accuracy and 98.4% face recognition accuracy while detecting abnormal behaviors such as face disappearance and identity

changes. Similarly, the AutoOEP framework [2] combines multiple techniques such as gaze tracking, head pose estimation, and object detection to compute cheating probability in real time. These studies demonstrate a shift from simple identity verification to continuous behavioral monitoring using multiple visual indicators.

Other studies further support the use of automated proctoring. The Talent-Interview system [3] performs cheating detection directly on the client side using face recognition, object detection, and voice monitoring. In addition, AI-based proctoring research [4] shows that systems can automatically flag suspicious behavior and record evidence for review by human proctors. Another study reports 94.2% facial verification accuracy and 89.7% head movement detection accuracy using computer vision techniques [5]. These findings suggest that AI can improve monitoring efficiency while assisting human supervision rather than replacing it.

Advancements in computer vision models have also improved real-time monitoring capabilities. YOLO-based systems [6] enable object detection models to identify facial orientation, mouth movement, and prohibited objects such as mobile phones during exams. Furthermore, multi-perspective monitoring approaches [7] show that using multiple cameras and gaze recognition improves detection accuracy and provides better analysis of examinee behavior. These developments indicate that modern systems focus not only on presence detection but also on understanding behavioral patterns

Despite these advancements, many institutions still use partially integrated systems. Based on environmental scanning, Legacy College of Compostela (LCC) currently does not utilize a computer-based examination system with integrated face

tracking and identity verification. However, detection of suspicious behavior still depends on CCTV monitoring or assigned proctors. During large-scale examinations, this approach makes it difficult to monitor all examinees consistently, which may lead to missed suspicious behaviors and increased workload for proctors.

In addition, many existing systems focus mainly on exam administration, such as question management and automated scoring, while providing only passive monitoring features like webcam recording or screen locking. These systems do not analyze behavior in real time, limiting their ability to detect actions related to gaze direction, posture, or facial movement. As a result, institutions still rely heavily on manual review processes.

Given these limitations, there is a need for an integrated system that combines examination management and automated behavioral monitoring. Existing solutions are often fragmented, separating exam delivery from monitoring tools, which reduces efficiency and limits analytical support.

To address this gap, this study proposes Knowing Eye: A Web-Based Examination Platform with Behavior Monitoring Using Facial and Postural Analysis. The system integrates exam creation, administration, and monitoring into a single platform. It uses computer vision and deep learning techniques such as YOLO for detection, convolutional neural networks (CNN) for feature extraction, and FaceNet or ArcFace for facial verification. The system analyzes head movement, eye gaze, and posture in real time and generates behavioral reports to assist administrators in evaluating exam integrity.

This study supports the Commission on Higher Education (CHED) ACHIEVE Framework by promoting digital innovation and the use of emerging technologies to address real-world challenges. By integrating artificial intelligence and web-based systems, the study aims to improve efficiency, transparency, and reliability in modern examination environments.

For clarity, key terms are defined in this study. Online examination platforms refer to systems used to create and manage exams digitally. Computer vision refers to techniques that allow computers to interpret visual data from images or video. Behavioral monitoring refers to the analysis of examinee actions such as gaze, posture, and facial movement. Deep learning refers to neural network-based methods used for pattern recognition and classification.

Overall, this study responds to the need for more reliable and efficient online examination systems. By combining automated behavioral monitoring with exam management, the proposed system aims to reduce reliance on manual supervision and support fair and consistent evaluation processes.

Objectives of the Study

General Objectives

To develop a Knowing Eye: A Web-Based Examination Platform with Behavior Monitoring Using Facial and Postural Analysis to enhance examination integrity, monitoring efficiency, and assessment reliability.

Specific Objectives

1. Develop a centralized web-based examination platform capable of managing exam creation, administration, and monitoring within a unified system.
2. Develop a monitoring and data management system that captures and stores examination-related data.
3. Apply computer vision techniques to analyze examinee behavior in real time in order to:

3.1 Detect facial presence and identity consistency

3.2 Track head movement, eye gaze, and posture

3.3 Identify abnormal or suspicious behavioral patterns

4. Integrate deep learning algorithms for behavioral analysis by:

4.1 Utilizing YOLO for face and posture detection

4.2 Applying convolutional neural networks (CNN) for feature extraction

4.3 Implementing FaceNet or ArcFace for facial feature consistency verification

5. Develop a behavior monitoring and scoring module that:

5.1 Flags potential anomalies during examination sessions

5.2 Generates behavioral indicators based on detected patterns

5.3 Supports review of flagged events by administrators

6. Generate an administrative dashboard that visualizes examination data and monitoring results to assist administrators in:

6.1 Reviewing examination sessions

6.2 Analyzing behavioral reports

6.3 Supporting decision-making regarding examination integrity

7. Evaluate the functionality, usability, and effectiveness of the developed platform in improving examination monitoring efficiency and supporting fair and reliable assessment processes.

Significance of the Study

This section presents the significance of the study by identifying its beneficiaries and explaining how the proposed system may contribute to improving examination:

Educational Institution. The proposed system helps improve examination integrity, monitoring efficiency, and assessment reliability through automated behavior monitoring and centralized examination management.

Examination Proctors. The system assists in monitoring examinees by detecting and recording suspicious behaviors, reducing the workload associated with manual supervision.

Instructors. The system provides additional behavioral reports that support fair and transparent evaluation of examination sessions.

Students. The system promotes a secure and fair examination environment by ensuring consistent monitoring during assessments.

Future Researchers. This study may serve as a reference for future researchers in creating web-based examination systems that utilize computer vision, artificial intelligence, and behavior monitoring technologies.

Scope and Limitation

Scope

The development of Knowing Eye: A Web-Based Examination Platform with Behavior Monitoring Using Facial and Postural Analysis. It aims to provide a centralized platform that enables examination creation, administration, and automated behavioral monitoring.

This study is conducted at Legacy College of Compostela (LCC) located on Dagohoy St. Poblacion, Compostela, Davao de Oro. Moreover, data utilized in this study consist of video streams captured through examinees' web cameras during examination sessions, system-generated behavioral logs, and examination interaction records. Publicly available datasets are used to support facial and posture detection

model training, while additional controlled datasets are collected through recorded mock examination sessions with participant consent. These datasets are used solely for model training, testing, and performance evaluation purposes.

Limitation

The study is limited for institutional examination environments where online or web-based implementation using desktop or laptop equipped with webcams. Mobile application deployment, integration with external institutional databases, biometric hardware devices, and nationwide operational deployment are beyond the scope of this research. Additionally, the system has no offline capabilities and It provides behavioral indicators rather than definitive accusations of cheating, with final judgment remaining under administrator discretion.

Review of Related Literature and Studies

This section presents the review of related literature and related studies relevant to the Web-Based Examination Platform with Behavior Monitoring Using Facial and Postural Analysis.

Related Literature

Artificial intelligence (AI) has become an important technology for improving the security and credibility of online examinations. According to Nigam et al. [8], AI-based proctoring systems employ computer vision and machine learning to automate identity verification and monitor examinee behavior throughout an assessment. Their review emphasized that facial recognition, gaze estimation, and head pose analysis are among

the most widely adopted techniques because they enable continuous monitoring while reducing the workload of human invigilators.

Building on these developments, Nafrees et al. [9] explained that deep learning models, particularly convolutional neural networks (CNNs), have significantly improved the accuracy of facial recognition compared with traditional image-processing methods. Their review found that CNN-based approaches provide reliable face detection even under varying lighting conditions and facial orientations, making them suitable for real-time online examination systems.

As online assessments become more common, researchers have also highlighted the importance of balancing examination security with ethical considerations. Han et al. [10] reported that digital proctoring technologies have strengthened academic integrity in higher education through automated identity verification and behavioral monitoring. However, the authors stressed that developers should ensure transparency, data privacy, fairness, and accessibility when implementing AI-powered examination systems.

Beyond identity verification, intelligent proctoring systems increasingly rely on behavioral analysis to identify suspicious activities during examinations. Kaddoura and Gumaï [11] proposed that integrating facial recognition with head pose estimation and eye gaze tracking improves cheating detection because multiple behavioral indicators provide more reliable evidence than a single monitoring technique. Their work demonstrated that multimodal computer vision significantly enhances the performance of automated examination monitoring.

Student posture has also emerged as an important behavioral indicator in online assessments. Shrivastava and Sonawane [12] demonstrated that combining body posture analysis with facial monitoring allows intelligent systems to recognize abnormal movements that may indicate suspicious behavior. Their findings showed that multimodal behavioral analysis produces more accurate monitoring results than facial recognition alone.

Similarly, Jadhav and Mehta [13] emphasized the growing importance of eye gaze estimation in AI-powered online examination platforms. Their review showed that deep learning-based gaze tracking enables systems to determine whether students maintain visual attention on the examination screen, making it an effective indicator for detecting potential cheating behavior while maintaining real-time performance.

While technical improvements continue to enhance monitoring accuracy, ethical concerns remain an important consideration. Miranda et al. [14] argued that AI-based educational surveillance systems should incorporate privacy-preserving mechanisms, explainable artificial intelligence, and responsible data management to ensure that automated proctoring remains fair, transparent, and acceptable to students and educational institutions.

Finally, Holden et al. [15] concluded that technology alone cannot completely eliminate academic dishonesty but can substantially strengthen examination integrity when integrated with institutional policies and appropriate assessment practices. Their review suggested that combining intelligent monitoring technologies with sound

academic regulations provides the most effective approach to maintaining fairness in online examinations.

Related Studies

Recent studies have demonstrated the effectiveness of artificial intelligence in improving the security of online examination systems through automated behavior monitoring. Tiong and Lee [16] developed an intelligent online examination proctoring system that employed deep learning models to detect suspicious behaviors during remote examinations. Their evaluation showed that the DenseLSTM model achieved approximately 95% detection accuracy, indicating that deep learning can effectively reduce academic dishonesty while minimizing the need for continuous human supervision.

To improve the efficiency of reviewing examination recordings, Li et al. [17] introduced a visual analytics framework that automatically analyzes webcam videos and user interactions during online assessments. Instead of manually reviewing lengthy recordings, instructors were provided with summarized behavioral events and highlighted suspicious activities. The study demonstrated that visual analytics can significantly reduce review time while improving the identification of possible examination violations.

Recognizing the limitations of single-feature monitoring, Noorbehbahani et al. [18] investigated various cheating detection approaches used in online examinations. Their study concluded that combining facial recognition, eye gaze tracking, browser activity monitoring, and identity verification provides higher reliability than relying on a

single behavioral indicator. The researchers recommended adopting multimodal monitoring frameworks to strengthen examination integrity.

Similarly, Singh and Das [19] developed a computer vision-based cheating detection system that combined eye gaze estimation with head pose analysis. Experimental results showed that integrating these two behavioral features improved the detection of suspicious activities, such as repeatedly looking away from the examination screen and unusual head movements. The study demonstrated that multiple behavioral indicators provide more accurate monitoring than gaze tracking alone.

Further improvements were presented by Alkhalisy et al. [20], who combined Dlib facial landmark detection with YOLO object detection to recognize abnormal student behaviors during online examinations. Their system detected facial orientation, eye movements, and prohibited objects in real time, resulting in more accurate behavioral assessment while maintaining efficient processing performance suitable for web-based environments.

Advances in multimodal artificial intelligence were further demonstrated by Bhogate et al. [21], who designed an AI-based examination proctoring system integrating face detection, eye tracking, head pose estimation, and mouth movement analysis. Their implementation showed that combining multiple behavioral indicators increased monitoring reliability and reduced false alarms compared with conventional webcam monitoring approaches.

Machine learning has also been applied to specialized online assessments. Ortín et al. [22] developed a deep learning assistant for online programming examinations that analyzed screenshot sequences using convolutional and recurrent neural networks. Their framework achieved a classification accuracy of more than 95%, demonstrating the effectiveness of AI-assisted monitoring in identifying unauthorized activities during programming examinations.

DER et al. [23] proposed a multimodal online examination monitoring framework that combined facial recognition, gaze estimation, posture analysis, and object detection into a unified system. Experimental evaluation showed that integrating these behavioral indicators enabled the system to detect suspicious activities with high accuracy while maintaining low false-positive rates.

Recent work has also explored lightweight intelligent monitoring suitable for real-time deployment. An AI-driven cheating detection framework [24] integrated eye movement analysis, head pose estimation, and YOLOv8-based mobile phone detection to generate immediate alerts whenever suspicious behaviors were identified. The study demonstrated that combining object detection with behavioral analysis improved the robustness of automated proctoring systems.

A more comprehensive multimodal framework was introduced by Naveen et al. [25] through the AutoOEP system. The proposed framework combined ArcFace identity verification, gaze estimation, head pose analysis, mouth movement detection, and YOLO-based prohibited object recognition, while an LSTM model calculated the probability of cheating in real time. The system achieved an overall classification

accuracy of approximately 90.7%, demonstrating that integrating multiple artificial intelligence techniques provides an effective solution for maintaining academic integrity in online examinations.

Furthermore, these studies demonstrate that intelligent examination platforms have evolved from simple webcam monitoring into comprehensive multimodal systems capable of analyzing facial features, gaze direction, head movements, posture, and prohibited objects.

Synthesis

The reviewed literature and related studies indicate that artificial intelligence and computer vision have significantly improved the security and reliability of online examination systems. Technologies such as facial recognition, eye gaze estimation, head pose detection, posture analysis, and object detection have been shown to effectively monitor examinee behavior and support academic integrity during remote assessments. Furthermore, studies demonstrate that combining multiple behavioral indicators provides more accurate and reliable monitoring than using a single detection method.

Despite these advancements, existing systems often focus on selected monitoring features or require complex deployment environments. There remains a need for an accessible web-based examination platform that integrates facial and postural analysis into a unified behavior monitoring system capable of operating in real time using standard webcams.

This study aims to develop a Web-Based Examination Platform with Behavior Monitoring Using Facial and Postural Analysis. By integrating facial and postural analysis within a web-based platform, the system is expected to enhance examination integrity, assist instructors in monitoring examinees more effectively, and provide a practical solution for secure online assessments.

Definition of Terms

ArcFace.	A deep learning-based facial recognition model that generates facial embeddings for identity verification by comparing the similarity between facial features.
Behavioral Monitoring.	The process of observing and analyzing an examinee's behavior, including facial movements, gaze direction, and body posture, during an online examination to identify possible irregularities.
Computer Vision.	A field of artificial intelligence that enables computers to acquire, process, and interpret visual information from images or video.
Deep Learning.	A subset of artificial intelligence that utilizes multilayer neural networks to learn patterns from data for tasks such as image recognition, classification, and facial analysis.
Online Examination Platform.	A web-based system used to create, administer, monitor, and evaluate examinations through digital technologies.
Session.	A single examination attempt that begins when an examinee starts an exam and ends upon submission, expiration, or termination of the examination.
WebSocket.	A full-duplex protocol for streaming frames and alerts in real time.
YOLO (You Only Look Once).	A real-time object detection algorithm used to identify and locate objects, such as persons or prohibited items, within video frames during an online examination.

CHAPTER 2

METHODOLOGY

This chapter presents the methodology utilized in the development of the proposed system entitled “Knowing Eye: A Web-Based Examination Platform with Behavior Monitoring Using Facial and Postural Analysis.” The study adopted the Agile Software Development Methodology as the framework for system development. Agile is a lightweight and adaptive software development approach that promotes continuous improvement, stakeholder collaboration, iterative development, and rapid response to changing requirements. Unlike traditional development methods, Agile focuses on delivering functional software in incremental stages while allowing modifications throughout the development process[#].

Figure # illustrates the Agile Software Development Methodology adopted in this study. The methodology follows the major phases of the System Development Life Cycle (SDLC), namely Planning, Design, Development, Testing, Deployment, and Maintenance. Each phase contributes to the successful development and continuous improvement of the proposed examination platform.

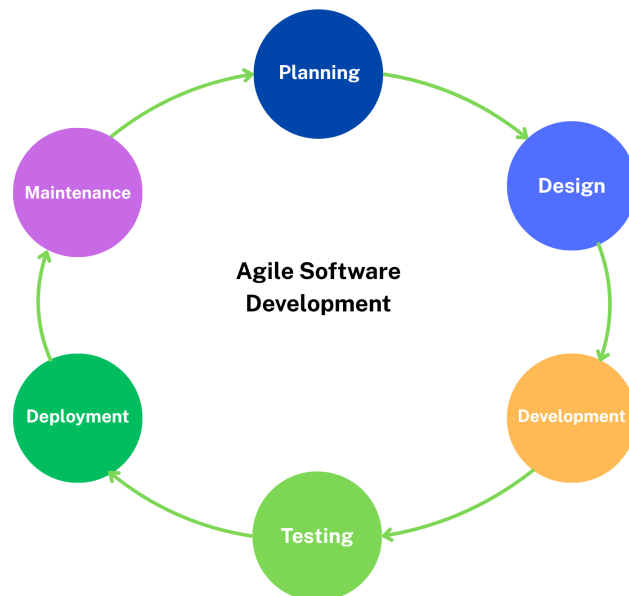


Figure 1. Agile Software Development Methodology

System Planning

The researchers presented three proposed capstone titles to the panel during the preliminary defense. After evaluating the proposed titles, the panel approved "Knowing Eye: A Web-Based Examination Platform with Behavior Monitoring Using Facial and Postural Analysis" as the official capstone project.

The researchers conducted environmental scanning on June 1, 2026 at Legacy College of Compostela, Compostela, Davao de Oro. The environmental scanning was conducted to gather information regarding the institution's existing examination process, identify the challenges encountered during online assessments, and determine the requirements for implementing a web-based examination platform with behavior monitoring. The information gathered served as the basis for the design and development of the proposed system, which aims to improve the integrity and reliability of online examinations through facial and postural analysis.

Project Team Organization

Each member of the project development team is assigned specific roles and responsibilities that contribute to the successful completion of the project.

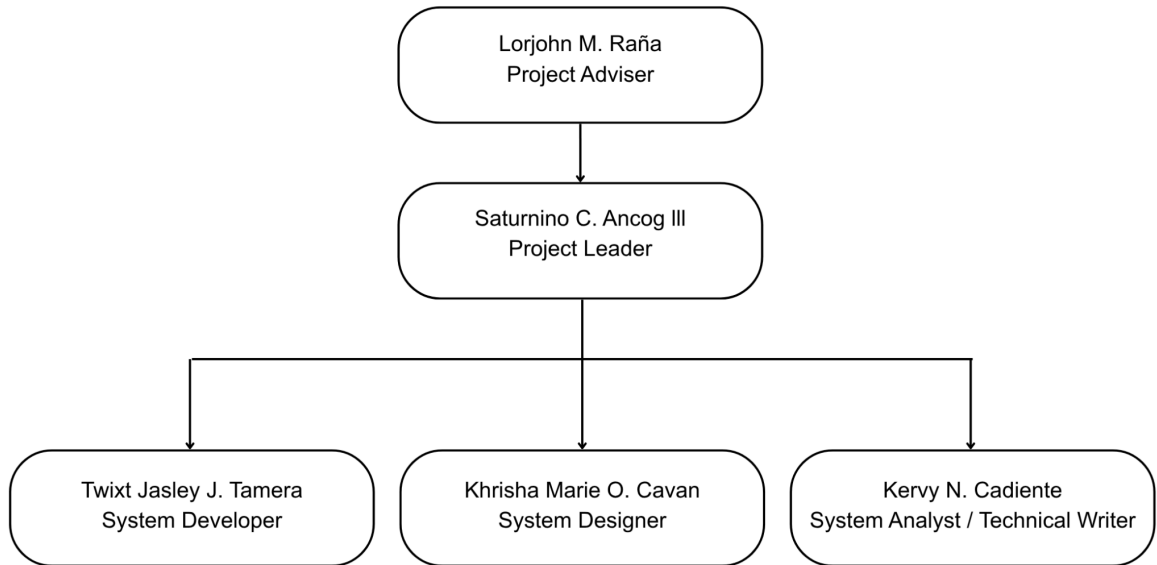


Figure 2. Chart of the Project Development Team

Work Breakdown Structure

This section presents the Work Breakdown Structure (WBS) of the proposed system. It outlines the major phases and tasks involved in the development of the Knowing Eye Examination Platform. Figure 3 illustrates the project divided into manageable components to ensure an organized and systematic development process.

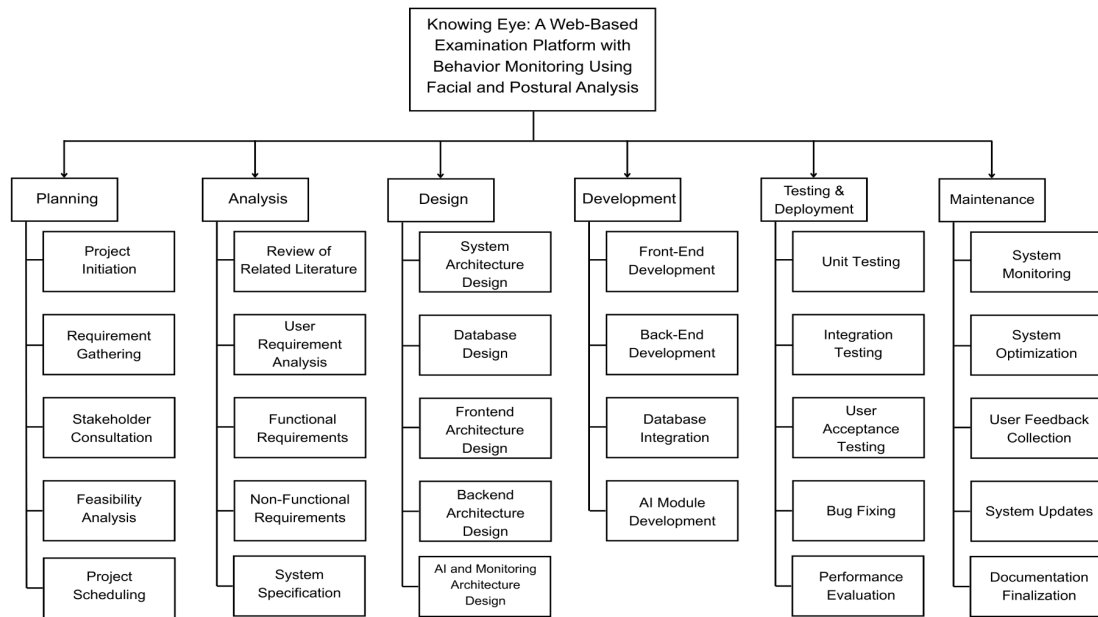


Figure 3. Work Breakdown Structure of the Project

The project follows the PADIM-based development framework consisting of Planning, Analysis, Design, Implementation, and Maintenance phases. Each phase contains specific activities necessary for the completion of the proposed system.

Gantt Chart

This section presents the project schedule for the development of Knowing Eye: A Web-Based Examination Platform with Behavior Monitoring Using Facial and Postural Analysis, as listed in the Work Breakdown Structure. A Gantt Chart is a graphical representation of the project timeline that illustrates the duration and sequence of activities throughout the development process. Figure # shows the schedule of major project activities from planning to maintenance.



Figure 4. Gantt Chart Project Planning

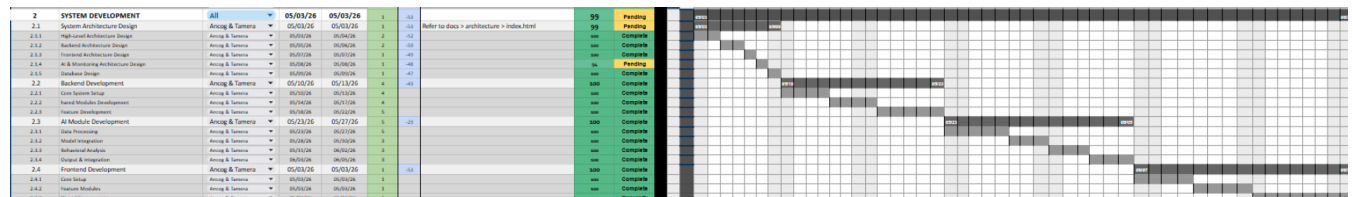


Figure 5. Gantt Chart System Development

System Analysis

After conducting environmental scanning, the researchers identified the requirements needed for the development of the proposed system. The system is designed to provide an online examination platform with real-time behavior monitoring through facial and postural analysis. It enables examinees to take examinations online while allowing administrators to manage examinations and monitor user activities.

System Architecture

Figure 6 presents the system architecture of the proposed web-based examination platform. The system consists of the Client Tier, Server Tier, and Data Tier.

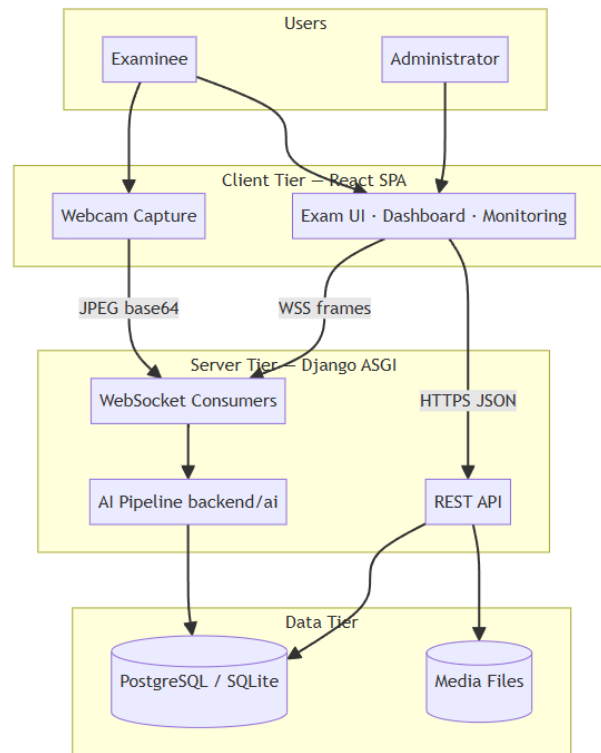


Figure 6 : Conceptual System Architecture

Functional and Non-Functional Requirements

Requirements were derived from capstone objectives, stakeholder interview and iterative prototype validation. Each item is numbered for traceability to test cases (prefixes FR-, NFR-).

Function Requirements

1. The system shall allow users to register and authenticate using username/password with JWT access and refresh tokens.

2. The system shall enforce role-based access control with two roles: Administrator and Examinee.
3. Administrators shall create, update, publish, archive, and delete exams and associated questions.
4. Administrators shall manage user accounts (activate, deactivate, assign roles).
5. Examinees shall start an exam session for a published exam within its availability window and attempt limit.
6. The system shall enforce one active session per examinee per exam at a time.
7. Examinees shall answer questions within a timed session and submit responses for automatic scoring.
8. The system shall calculate percentage score and pass/fail based on exam passing threshold.
9. Examinees shall enroll a reference face image before or during monitoring for identity verification.
10. The system shall capture webcam frames and analyze behavior in near real time via WebSocket (REST fallback).
11. The AI pipeline shall detect face absence, multiple faces, gaze deviation, poor posture, prohibited objects, and identity mismatch.
12. The system shall persist behavior events and generate severity-classified alerts linked to the session.
13. Administrators shall view live alerts and resolve them through the behavior API.

14. Administrators shall access dashboard KPIs, session reports, timeseries analytics, and CSV export.

15. Examinees shall view their own session history and results where authorized.

Non-Function Requirements

1. Performance: Frame analysis target ≤ 300 ms per frame at $\sim 1\text{--}5$ FPS; REST API responses ≤ 2 s under normal load.
2. Scalability: Support multiple concurrent examinees on a single server; production deployment uses Redis channel layer for multi-worker WebSocket broadcast.
3. Security: Passwords hashed (Django PBKDF2); JWT for API/WS auth; HTTPS/WSS in production; RBAC on all endpoints.
4. Reliability: AI adapter degrades to deterministic stub when ML dependencies unavailable; monitoring API contract always honored.
5. Usability: Responsive web UI; role-specific navigation; clear exam timer and compliance feedback during monitoring.
6. Maintainability: Feature-based modular monolith; documented REST catalog; 37 automated backend tests.
7. Portability: Runs on Windows/Linux; SQLite for development; PostgreSQL for production without code changes.
8. Privacy: Raw video frames are not stored in the database by default (KE_STORE_FRAMES=False); only derived metrics and events persist.

Use Case Diagram

A Use Case Diagram presents the interactions between the users and the Knowing Eye System. As shown in Figure #, the system has two main actors: the Administrator and the Examinee.

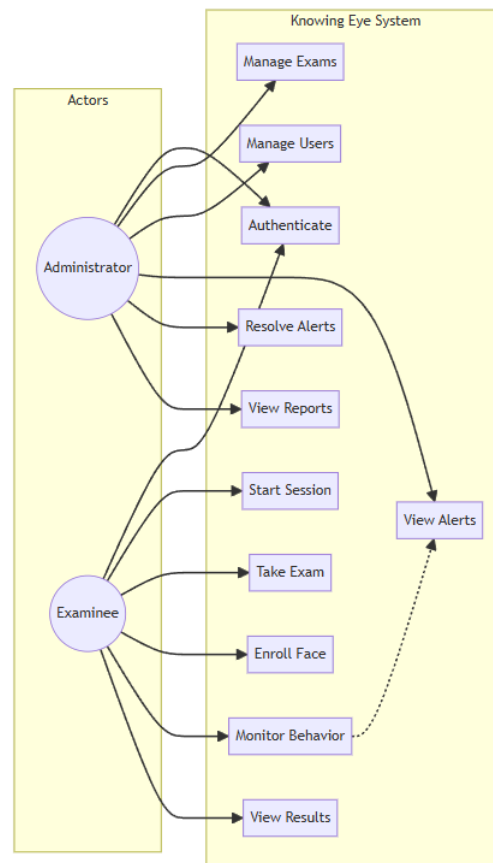


Figure 7: Use Case Diagram

Context Flow Diagram (Level 0 DFD)

The researchers have used the Context Flow Diagram (CFD) to show the data exchange between the Administrator, Examinee, Webcam Device, and the Knowing Eye Examination Platform. Figure 8 presents the inputs and outputs of the system, including exam data, credentials, answers, video frames, alerts, reports, scores, and feedback.

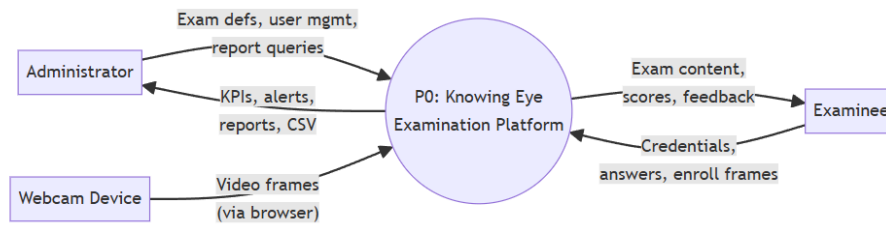


Figure 9: ERD Level 0

Level 1 DFD

The researchers have used the Level 1 Data Flow Diagram (DFD) to show the detailed processes of the system. Figure 10 presents authentication, exam management, exam sessions, behavior analysis, alert recording, and report generation, including the related data stores.

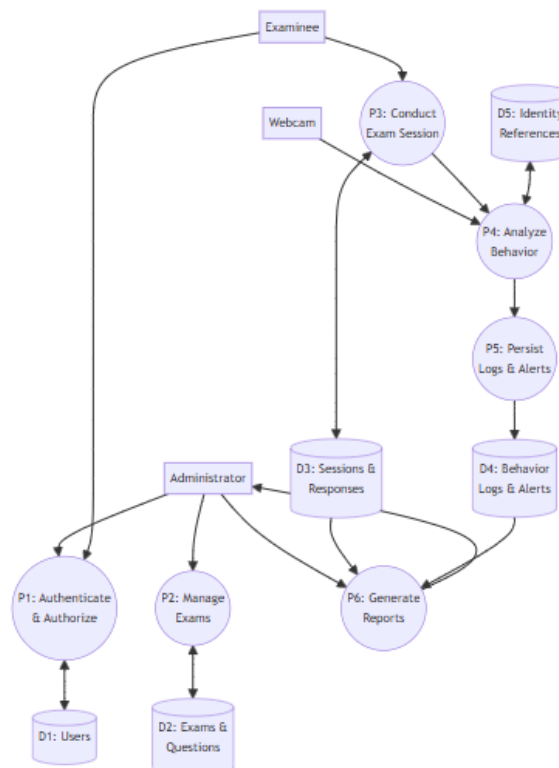


Figure 10: ERD Level 1

System Design

This section discusses the planned system design, including the database structure and relationships of the proposed system.

The researchers have used the Entity Relationship Diagram (ERD) to show the logical structure and relationships of the system database. Figure 11 presents the entities for users, examinations, questions, exam sessions, responses, session logs, behavior logs, behavior alerts, and identity references.

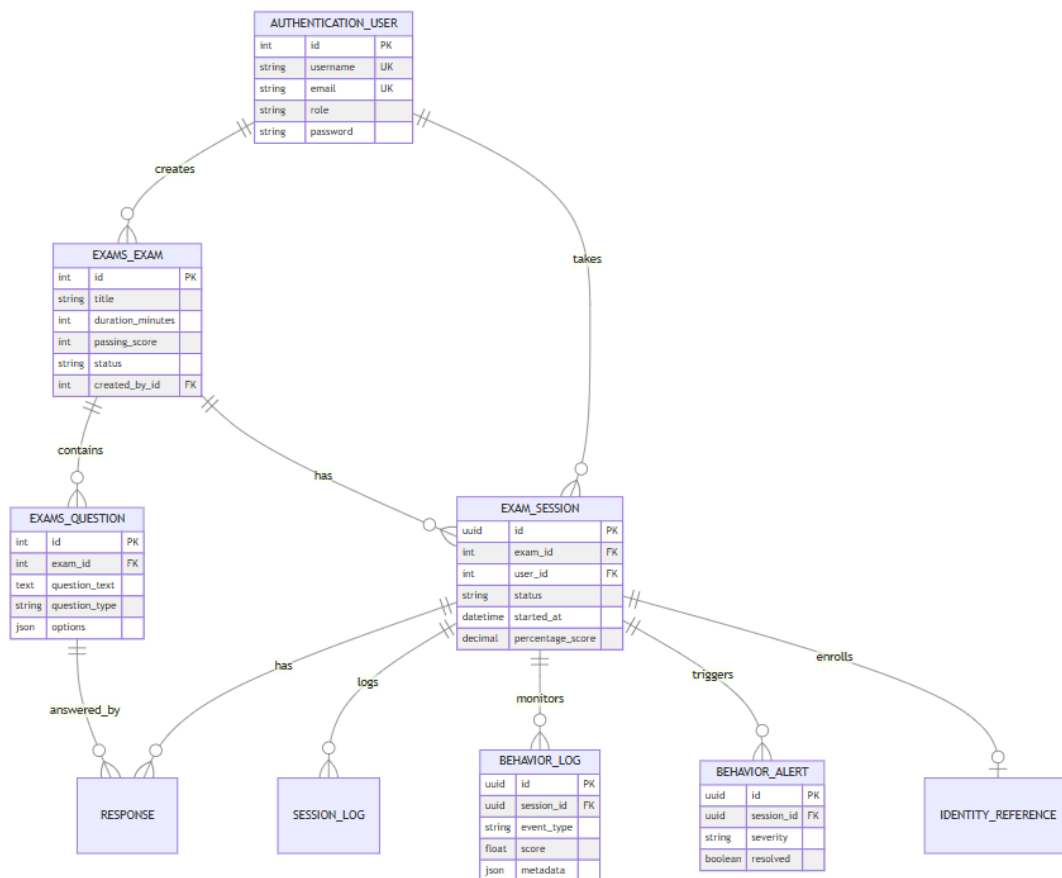


Figure 11: Entity Relationship Diagram

Data Dictionary

The researchers have used the Data Dictionary to provide a complete listing of the primary tables, fields, data types, constraints, and descriptions used in the Knowing Eye Examination Platform. It defines the data stored for user accounts, examinations, questions, exam sessions, responses, monitoring logs, alerts, and identity references.

Table 1. Authentication_user

Field	Type	Constraints	Description
id	INTEGER	PK,auto	Unique user identifier
username	VARCHAR(150)	UNIQUE, NOT NULL	Login username
email	VARCHAR(254)	UNIQUE, NOT NULL	Contact email
password	VARCHAR(128)	NOT NULL	Hashed password (PBKDF2)
first_name, last_name	VARCHAR(150)		Display name
role	VARCHAR(20)	ADMIN EXAMINEE	RBAC role
avatar	VARCHAR(100)	NULL	Profile image path
phone, institution, student_id	VARCHAR		Profile metadata
is_active	BOOLEAN	DEFAULT true	Account enabled flag
last_seen_at	DATETIME	NULL	Last activity timestamp
created_at, updated_at	DATETIME	auto	Audit timestamps

Table 2. Exams_exam

Field	Type	Constraints	Description
id	INTEGER	PK	Exam identifier
title	VARCHAR(255)	NOT NULL	Exam name
description, instructions	TEXT		Exam narrative and rules
duration_minutes	INTEGER	DEFAULT 120	Time limit
passing_score	INTEGER	0–100	Minimum % to pass
exam_code	VARCHAR(32)	UNIQUE, NULL	Optional access code
available_from, available_until	DATETIME	NULL	Scheduling window
max_attempts	INTEGER	DEFAULT 1	Attempts per user
status	VARCHAR(20)	draft active archived	Lifecycle state
created_by_id	INTEGER	FK → user	Authoring administrator

Table 3. exams_question

Field	Type	Constraints	Description
id	INTEGER	PK	Question identifier
exam_id	INTEGER	FK → exam	Parent exam
question_text	TEXT	NOT NULL	Prompt text
question_type	VARCHAR(20)	MCQ, T/F, etc.	Answer format
options	JSON	DEFAULT []	MCQ choices

correct_answer	TEXT	NOT NULL	Scoring key
points, order	INTEGER		Weight and display order

Table 4. Exam_sessions

Field	Type	Constraints	Description
id	UUID	PK	Session identifier (used in WebSocket URL)
exam_id, user_id	INTEGER	FK	Exam and examinee
status	VARCHAR(20)	in_progress completed terminated expired	Session state
started_at, submitted_at	DATETIME		Session timeline
time_remaining	INTEGER		Seconds left at submit
total_score, percentage_score	INT / DECIMAL	NULL until submit	Computed results
passed	BOOLEAN	NULL until submit	Pass/fail flag
ip_address, user_agent	VARCHAR / TEXT		Client audit trail

Table 5. Response, session_logs, behavior_alert, monitoring_identity_reference

Table	Key Fields	Description
responses	session_id, question_id, answer_text, is_correct, time_spent	Per-question answers submitted during session

session_logs	session_id, event_type, details, timestamp	Lifecycle audit (started, submitted, terminated)
behavior_log	session_id, event_type, score, confidence, metadata	AI-detected behavioral events
behavior_alert	session_id, alert_type, severity, resolved	Flagged integrity violations
monitoring_identity_reference	session_id (1:1), embedding (JSON), backend	Enrolled face vector for identity verification

Technologies, Concepts, and Theories

The following were the technologies, concepts, and theories applied by the researchers to develop the Knowing Eye Examination Platform.

Visual Studio Code

Visual Studio Code is the primary integrated development environment (IDE) used by the development team. It provides a lightweight and extensible workspace for developing both the frontend and backend components of the web-based examination platform. Its built-in Git integration, debugging tools, terminal, and extensions support efficient development using Python, TypeScript, React, and Django. Figure 13 shows the Visual Studio Code IDE.

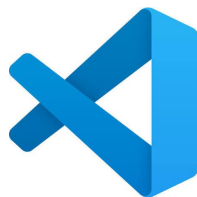


Figure 12: Visual Studio Code

Django 6, Django REST Framework, and Channels

Django, Django REST Framework, and Django Channels are used to develop the backend of the proposed system. Django is a high-level Python web framework that supports a feature-based modular monolith architecture, allowing modules such as user management, examinations, behavior monitoring, and reports to be organized in one maintainable application. It provides the Object-Relational Mapper (ORM), authentication mechanisms, database migration tools, and server-side validation. Django REST Framework is used to create RESTful application programming interfaces (APIs) that connect the frontend interface with backend services, including user authentication, examination records, monitoring data, and generated reports. JSON Web Token (JWT) authentication is implemented to secure user sessions and role-based access. Django Channels extends Django by enabling asynchronous communication through WebSockets, allowing the system to transmit webcam analysis results, alerts, and examination events in real time between the examinee interface and the monitoring dashboard. Figure 14 shows Django.



Figure 13: Django

React 18 and Vite

React and Vite are used to develop the frontend interface of the proposed system. React is used to build the single-page application through reusable components for the examination-taking page, live monitoring dashboard, administrator controls, and reporting views. It also supports protected routing, ensuring that users can only access system functions based on their assigned roles, such as administrator, instructor, proctor, or examinee. Vite serves as the development and build tool for the React application by providing fast project startup, hot module replacement, and optimized production builds. These features improve the efficiency of frontend development, testing, and deployment. Figure 15 shows the React and Vite .



Figure 14: React

TypeScript

TypeScript is used in the frontend development of the system to provide static typing for React components, API requests, monitoring data, and user roles. It helps identify possible errors during development before the application is executed.

TypeScript also improves code readability, maintainability, and consistency between the frontend and backend API contract. Figure 16 shows the TypeScript.



Figure 15: TypeScript

PostgreSQL

PostgreSQL is used for database management in the proposed system. PostgreSQL serves as the production relational database and stores user accounts, examination details, questions, answers, behavior-monitoring records, alerts, identity-verification results, and generated reports. It is selected because it provides strong data integrity, security, scalability, and support for complex database queries. Through the Django ORM, the database schema remains portable to PostgreSQL, allowing the development database to be transferred to the production environment with minimal changes. Figure 17 shows the PostgreSQL



YOLOv8

YOLOv8 is an object detection model used to identify persons and prohibited objects within webcam frames during online examinations. It can detect objects such as mobile phones that may indicate possible examination violations. The detected object-presence signals are included in the behavioral compliance score of the examinee. This helps the system identify suspicious activities and notify the assigned proctor in real time. Figure 18 shows the YOLOv8.



Figure 17: YOLOv8

MediaPipe

MediaPipe is used for facial landmark detection and body posture analysis. Its face and pose solutions extract facial landmarks and body keypoints from webcam frames. These landmarks are used to estimate gaze direction, head orientation, face visibility, and posture behavior during an examination. The resulting information contributes to the system's behavioral monitoring process. Figure 19 shows the MediaPipe.



Figure 18: MediaPipe

ArcFace and InsightFace

ArcFace and InsightFace are used for examinee identity verification. ArcFace generates 512-dimensional facial embeddings using angular-margin learning, while InsightFace provides the tools needed to process and compare facial data. The system compares the live webcam facial embedding with the enrolled reference image through cosine-distance matching. If the similarity score falls below the acceptable threshold, the system flags a possible identity mismatch for proctor review. Figure 20 shows the ArcFace and InsightFace.



Figure 19: PostgreSQL

System Testing

This section of the document presents the system testing and implementation phase of the Knowing Eye Examination Platform. The results are pending formal User Acceptance Testing (UAT) and AI benchmark evaluation. The tables below will be completed after the testing activities are conducted.

Testing Approach

The researchers will use a layered testing approach based on IEEE test documentation and automated Django tests. The testing will cover unit and API testing, integration testing, WebSocket testing, AI pipeline testing, and user acceptance testing.

Table 1. Results of System Testing

Test ID	Module / Feature	Expected Result	Observed Result	Status
ST-01	Authentication (login, JWT)	Valid credentials return tokens; invalid rejected		Pending
ST-02	Exam CRUD & publish	Admin creates/publishes; RBAC enforced		Pending
ST-03	Exam session & submit	Session starts, answers scored		Pending
ST-04	Monitoring WebSocket	Frames accepted; alerts broadcast		Pending
ST-05	Behavior logs & reports	Logs queryable; CSV export works		Pending
ST-06	AI pipeline output	Valid schema; flags on blank frame		Pending

Table 2. Classification Metrics (Accuracy, Recall, Precision, F1-Score)

Metric	Target	Observed	Interpretation	Status
Accuracy	≥ 90%		Overall correct classifications	Pending
Recall	≥ 88%		Sensitivity to true violations	Pending
Precision	≥ 88%		Few false alarms	Pending
F1-Score	≥ 88%		Balance of precision and recall	Pending

Testing Results

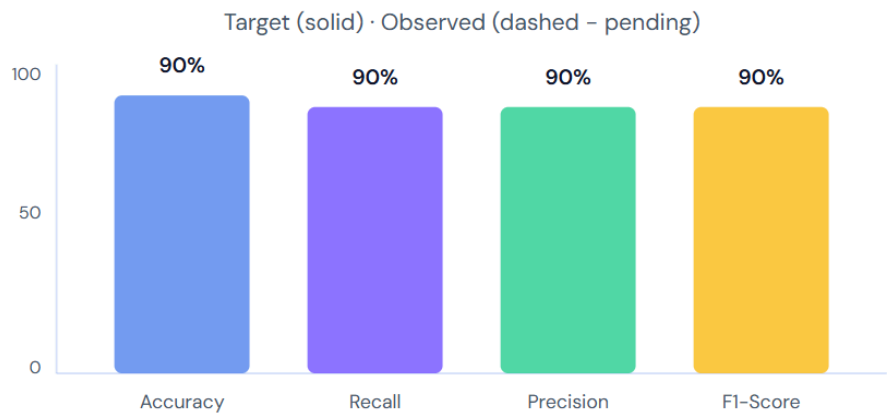


Figure 20: Entity Relationship Diagram

Figure 20 illustrates the target classification metrics for the behavior-monitoring component of the Knowing Eye Examination Platform. The figure presents the target values for accuracy, recall, precision, and F1-score. The target accuracy is set at 90%,

while recall, precision, and F1-score are each set at 88%. The observed results are pending and will be included after the formal testing and evaluation of the AI pipeline.

Summary of Findings

Req ID	Requirement	Method	Result	Remarks
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Researchers have prepared the Summary of Findings to present the results of the system testing based on the requirements of the Knowing Eye Examination Platform. The table includes the requirement identification, requirement description, testing method, result, and remarks. The traceability matrix will be completed after the conduct of User Acceptance Testing (UAT).

System Maintenance

This section will describe the post-deployment sustainment of the Knowing Eye platform, ensuring that the system remains secure, available, and adaptable throughout its operational lifecycle. Rather than treating deployment as the endpoint of the project, this phase will treat it as the beginning of a continuous cycle of monitoring, patching, and improvement. The plan will be organized into two complementary efforts: a systems security plan that safeguards examinee privacy, exam integrity, and institutional data, and a systems maintenance plan that preserves availability while evolving the platform in response to user feedback and changing institutional policy. All controls and activities are stated as planned actions that will be implemented and validated during the operational phase following deployment.

Systems Security Plan

The system will adopt a defense-in-depth security posture that layers preventive, detective, and corrective controls across the authentication, transport, application, and data tiers so that no single point of failure can compromise the platform. At the authentication and authorization layer, user credentials will be protected using Django's PBKDF2 password hashing with an enforced minimum length and rotation upon any suspected breach, while access will be governed by JSON Web Tokens consisting of a sixty-minute access token and a seven-day refresh token with rotation and blacklisting enabled. Role-based access control will distinguish ADMIN and EXAMINEE roles to guard every API and WebSocket route, and WebSocket connections will be authenticated through a JWT validated by the JWTAuthMiddleware and carried exclusively over secure WSS in production. New endpoints will be routinely audited to confirm that the correct permission classes are attached, and tokens will be revoked immediately whenever a compromise is detected. At the transport and network layer, all production traffic will be encrypted using TLS 1.2 or higher through Nginx, with SECURE_SSL_REDIRECT and HTTP Strict Transport Security enabled whenever DEBUG is set to false. Cross-origin requests will be limited to known frontend origins through CORS_ALLOWED_ORIGINS, request volume will be constrained by DRF throttling at sixty requests per minute for anonymous users and twelve hundred per minute for authenticated users, and the combination of ALLOWED_HOSTS and the AllowedHostsOriginValidator will defend against host-header attacks. At the application and data layer, each sensitive asset will be matched to a specific mitigation: exam content will be protected by JWT and ownership checks with draft exams hidden from examinees; behavior logs will be session-scoped so that examinees can view only their

own data; face embeddings will be stored on a per-session basis and deleted together with the session to prevent biometric misuse; media uploads such as avatars will undergo size and type validation; and application secrets will remain gitignored, with a strong `DJANGO_SECRET_KEY` required in production. Finally, security monitoring and incident response will be sustained by regularly reviewing `backend/logs/known_eye.log` for spikes in 401, 403, and 500 responses, by tracking unresolved behavior alerts through the admin dashboard, and by following a defined breach procedure in which the team will rotate the `DJANGO_SECRET_KEY`, invalidate the JWT blacklist, and force password resets for affected accounts. Security patches will be applied monthly using `pip audit` and `npm audit`.

Systems Maintenance Plan

Keeping the platform healthy after launch will rely on four kinds of maintenance that work together: corrective, adaptive, perfective, and preventive. Each one keeps the system available, patched, and in step with what users and the institution actually need. Corrective maintenance will happen whenever something breaks. The development team will track bugs that surface through logs, user reports, and failed tests, then push out hotfixes to resolve them. Adaptive maintenance will take place every semester, when administrators and developers refresh exam schedules, question banks, passing thresholds, and the `pipeline.yaml` tuning. Perfective maintenance will run on a quarterly basis to bring performance tuning, UI refinement, better reports, and upgrades to the YOLO and ArcFace models. Preventive maintenance will be handled weekly and monthly through database backups, dependency updates, disk cleanup, and log rotation so small problems never get the chance to grow. Behind these categories sit the

day-to-day procedures that keep everything running. The PostgreSQL database will be backed up every day with `pg_dump` to secure storage, kept on a thirty-day rolling retention window, and test-restored each quarter to make sure the backups actually work. Every release will run `migrate` and `collect static` before going live, after which the Daphne workers will be restarted and the frontend rebuilt with `npm run build`. For the AI pipeline, the team will adjust the `pipeline.yaml` thresholds to match proctoring policy and re-run `python manage.py test ai.tests` after any change to the computer-vision dependencies. Logs and storage will be kept tidy with a rotating log of five megabytes across five backups, and old sessions will be archived according to the institution's retention policy. Configuration will stay cleanly separated between development and production so nothing risky leaks into a live environment. In production, `DJANGO_DEBUG` will be turned off, the `DJANGO_SECRET_KEY` will be a strong random string of fifty or more characters, PostgreSQL will take over from SQLite, Redis will back Channels, and frame storage will stay disabled by default to protect examinee privacy. Support will follow a simple tiered model based on how badly an issue hurts. Critical incidents such as a platform outage or mass login failure will be restored within four hours, high-severity problems like an offline monitoring pipeline within one business day, medium-severity defects within three business days, and low-severity cosmetic or documentation items in the next release cycle.

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